COMSATS UNIVERSITY ISLAMABAD

ATTOCK CAMPUS

Computer Science Department

Course Title: Information Security

Assighment 1

Submitted by:

Mehak (FA24-BSE-007)

Submitted to:

Ms. Ambreen Gul

1. Code:

```python
# Caesar Cipher Implementation - Modified Version

def encrypt_message(message, key):  1 usage
    result = ""

    for ch in message:
        if ch.isalpha():  # Check if character is a letter
            ascii_base = 65 if ch.isupper() else 97
            result += chr((ord(ch) - ascii_base + key) % 26 + ascii_base)
        else:
            result += ch   # Keep special characters unchanged

    return result


def decrypt_message(secret_text, key):  1 usage
    original = ""

    for ch in secret_text:
        if ch.isalpha():
            ascii_base = 65 if ch.isupper() else 97
            original += chr((ord(ch) - ascii_base - key) % 26 + ascii_base)
        else:
            original += ch

    return original
```

```python
def main():  1 usage
    user_text = input("Type your message: ")
    shift_value = int(input("Enter shift number: "))

    cipher_text = encrypt_message(user_text, shift_value)
    print("Cipher Text:", cipher_text)

    plain_text = decrypt_message(cipher_text, shift_value)
    print("Original Text:", plain_text)


if __name__ == "__main__":
    main()
```

2. Line-by-Line Explanation

Encryption Function

- encrypt_message(message, key)
  This function encrypts the input message using the Caesar Cipher method.
- result = ""
  An empty string is created to store encrypted text.
- for ch in message:
  Loops through each character in the message.
- if ch.isalpha():
  Checks whether the character is a letter.
- ascii_base = 65 if ch.isupper() else 97
  Determines ASCII base value depending on uppercase or lowercase.
- chr((ord(ch) - ascii_base + key) % 26 + ascii_base)
  Converts the letter to ASCII, applies shift, wraps using modulo 26, and converts back to character.
- else:
  Keeps spaces and special characters unchanged.

Decryption Function

- decrypt_message(secret_text, key)
  This function reverses the encryption process.
- The only difference is subtracting the key instead of adding it.
- % 26 ensures the value stays within alphabet range.

Main Function

- Takes user input
- Calls encryption function
- Displays encrypted text
- Calls decryption function
- Displays original text

3. Output:

```
Type your message: my name is mehak
Enter shift number: 6
Cipher Text: se tgsk oy skngq
Original Text: my name is mehak

Process finished with exit code 0
```